

AL TANFITH AL DWALIAH SPC

LIVE QUIZ PLATFORM

Technical & Commercial Proposal

Prepared for: **Petroleum Development Oman (PDO)**

Reference: **ATD-LQP-2026-003**

Version: **1.0**

Date: **22/07/2026**

2. Document Control

2.1 Document Information

Title	Live Quiz Platform — Technical & Commercial Proposal
Reference	ATD-LQP-2026-003
Version	1.0
Date	22/07/2026
Prepared by	Haitham Al Lamki, Al Tanfith Al Dwaliah SPC

3. Table of Contents

2. Document Control.....	2
2.1 Document Information	2
3. Table of Contents	2
4. Letter of Submission	4
5. Executive Summary	5
6. About Al Tanfith Al Dwaliah SPC	6
7. Understanding of Client Requirements	7
8. Business Challenges	8
9. Proposed Solution	10
10. Platform Overview	10
11. Current Use Case	11
12. Functional Scope	12
12.1 Quiz Creation & Content.....	12
12.2 Live Hosting & Participation.....	13
12.3 Branding & Experience	14
12.4 Reporting & Analytics.....	14
12.5 Administration & Governance.....	14
12.6 AI Capabilities	15
13. AI Capabilities	15
13.1 Current Capabilities (Included).....	15

13.2 Optional Capabilities (Scoped per Engagement)	16
13.3 Future Roadmap	17
14. User Roles	17
15. Reporting and Analytics	18
16. Technical Architecture	20
16.1 Technology Stack	20
16.2 Architecture Diagram	20
17. Real-Time Architecture	21
17.1 Session Flow	21
17.2 Server-Authoritative Timing, Reconnection, and Why It Matters at Scale	21
18. Deployment Options	22
18.1 Cloud / SaaS	22
18.2 On-Premises	22
18.3 Hybrid	23
18.4 Supported Deployment Platforms	23
18.5 Client Standard Requirements	24
19. Cybersecurity	25
20. Data Protection	26
21. Data Ownership	26
22. Integration Options	27
23. Performance and Scalability	28
24. Implementation Methodology	29
25. Project Timeline	30
25.1 Milestones	31
26. Project Governance	31
27. Deliverables	32
28. Training and Knowledge Transfer	34
29. Testing and Acceptance	34
30. Support and Maintenance	35
31. SLA	35
32. Commercial Models	37
32.1 Model A — Annual SaaS Subscription	37
32.2 Model B — Perpetual Licence / One-Time Purchase	37
32.3 Model C — Enterprise Custom Development Partnership	37

32.4 Model Comparison	37
33. Pricing Approach	38
33.1 Pricing Methodology	38
34. Assumptions.....	38
35. Exclusions	39
36. Risk and Mitigation	39
37. Change Request Process	40
38. Intellectual Property	40
39. Proposal Validity.....	41
40. Payment Terms	41
41. Termination and Exit	41
42. Future Roadmap	41
42.1 Learning & Assessment.....	41
42.2 AI.....	41
42.3 Integrations.....	42
42.4 Platform	42
43. Why AI Tanfith AI Dwaliah SPC	42
44. Next Steps	42
45. Acceptance Page	43
46. Contact Details.....	43
47. Appendices.....	43
47.1 Appendix A — Glossary.....	43
47.2 Appendix B — Placeholder Index.....	44
47.3 Appendix C — Referenced Documents.....	44

4. Letter of Submission

Petroleum Development Oman (PDO)

22/07/2026

Dear Sir / Madam,

Subject: Proposal for the Supply, Configuration, and Support of the Live Quiz Platform

On behalf of AI Tanfith AI Dwaliah SPC ("Altanfith"), I would like to thank **Petroleum Development Oman (PDO)** for the opportunity to submit this proposal. We appreciate the time and consideration your team has invested in reviewing our solution, and we welcome the opportunity to support your organisation's engagement, training, and assessment objectives.

This proposal presents the Live Quiz Platform as an enterprise engagement, assessment, and analytics solution designed to help organisations such as **Petroleum Development Oman (PDO)** run live sessions, structured assessments, and knowledge-evaluation activities with the branding control, reporting depth, and security posture that a large organisation requires — going beyond what general-purpose public quiz tools are built to offer. The following sections set out our understanding of your requirements, the proposed solution, our technical architecture and delivery methodology, and the commercial models available for this engagement.

This proposal, including its commercial terms, is valid for a period of 30 days from the date shown above. We remain available throughout this period to answer questions, arrange a live demonstration, or refine any aspect of this proposal to better reflect **Petroleum Development Oman (PDO)**'s specific requirements.

We look forward to the possibility of working with **Petroleum Development Oman (PDO)** and would welcome the opportunity to discuss this proposal further at your convenience.

Yours faithfully,

Haitham Al Lamki

Al Tanfith Al Dwaliah SPC

Email: haitham@altanfith.com

Phone: +968 9937 1775

Sultanate of Oman

5. Executive Summary

Altanfith is pleased to present the Live Quiz Platform to **PDO** as an enterprise engagement, assessment, and knowledge-evaluation platform — a flexible, customisable alternative to general public quiz tools, purpose-built for organisations that require organisational branding, structured reporting, and enterprise-grade controls around how live sessions and assessments are run.

The platform is not a concept or a prototype. It is already in real operational use at Abraj Energy Services, where it supports HSE meetings and employee engagement activities. This production track record gives **PDO** confidence that the core platform is stable, functional, and capable of supporting real organisational events at meaningful scale, rather than a solution that exists only on paper.

The platform is bilingual by design, with a fully localised Arabic and English interface including complete right-to-left (RTL) support for Arabic — a capability that matters for organisations operating across Oman and the wider Gulf region, where sessions, dashboards, and reports need to be equally usable in either language.

Content creation is accelerated through AI-assisted generation: administrators can generate a complete quiz from a topic or title, or from an uploaded PDF, Word document, URL, or pasted text, with the AI assigning suggested difficulty levels and answer explanations. Every AI-generated item is presented to an administrator for review and editing before it is used in a live session — the platform is designed to keep a human reviewer in the loop rather than publishing AI output automatically. Additional AI capabilities, such as AI-assisted rewriting, suggestions, and generated visual themes, are available as optional, separately scoped additions, and a further set of AI-driven insights and reporting features sits on our future roadmap.

PDO can choose how the platform is hosted and delivered. Three deployment models are available: a fully managed cloud/SaaS option, a fully on-premises option in which the Express.js backend runs

entirely within the client's own infrastructure, and a hybrid option that splits components between the two — the right choice depends on **PDO**'s data residency, network, and cybersecurity requirements, and is confirmed through a dedicated infrastructure assessment rather than assumed in advance. Commercially, the platform is available under three models — an annual SaaS subscription, a one-time perpetual licence, and an enterprise custom development partnership — so that **PDO** can select the ownership and cost structure that best fits its procurement approach.

Underpinning all of this is the fact that Altanfith is an Omani company: delivery, support, and communication happen locally, in Arabic and English, without the delays or handover friction that come with an offshore vendor. This local presence supports the value of the platform — it is the foundation the offer is delivered on, not the offer itself.

Proposal at a glance	
Solution	Enterprise engagement, assessment & analytics platform — Live Quiz Platform
Proven use	In production at Abraj Energy Services — HSE meetings & employee engagement sessions
Deployment	Cloud/SaaS, fully on-premises, or hybrid — confirmed via infrastructure assessment
Commercial models	Annual SaaS subscription · perpetual licence · enterprise custom development partnership
AI	AI-assisted quiz generation from topic, PDF, document, URL, or text — always reviewed by an administrator before use
Support	Bronze / Silver / Gold packages, delivered locally from Oman in Arabic and English

6. About AI Tanfith AI Dwaliah SPC

AI Tanfith AI Dwaliah SPC is an Omani-owned SME (Small and Medium Enterprise). Altanfith is a Sultanate of Oman-based software company focused on designing, building, and supporting enterprise digital platforms for government, oil & gas, education, and private-sector organisations. Our work centres on delivering software that is not simply installed and left running, but actively configured, branded, and adapted around each client's operational, reporting, and security requirements.

Our core capability areas span custom application development, enterprise platform configuration, and applied artificial intelligence — including AI-assisted content generation of the kind used within the Live Quiz Platform itself. We combine this technical capability with a practical understanding of how large Omani organisations operate: the governance structures they answer to, the security review processes their IT and cybersecurity teams require, and the bilingual Arabic/English environment in which their staff and stakeholders work.

The Live Quiz Platform is a direct expression of this approach. Rather than offering a generic, one-size-fits-all quiz tool, we designed and continue to develop a platform that can be shaped around each

client's branding, workflows, integration landscape, and deployment preferences — and that is already proven in real operational use, as set out in Section 11 of this proposal.

As an Omani company, Altanfith is able to engage directly with **PDO**'s teams throughout discovery, delivery, and ongoing support, in Arabic and English, without reliance on an intermediary or an offshore support desk operating outside Gulf business hours.

7. Understanding of Client Requirements

Large Omani organisations — whether in government, oil & gas, education, or the private sector — typically approach an engagement and assessment platform with a common set of expectations that go well beyond what a free or public quiz tool is designed to provide. Based on our experience delivering the Live Quiz Platform into a live production environment, we understand these expectations to include the following.

Requirement	Our response
Full control over branding, so that sessions and materials reflect the organisation's own visual identity rather than a generic third-party look	Organisation branding, including logo and colour customisation, custom themes, custom backgrounds, and custom UI text, is built into the platform
Control over where data resides and how it is retained, including the ability to keep sensitive data within the organisation's own environment	A fully on-premises deployment option is available, in which the backend and database can run entirely within PDO 's own enterprise infrastructure environment; data retention and export terms are defined in the commercial agreement, and final data residency arrangements are confirmed during discovery and infrastructure assessment
A genuinely bilingual Arabic and English experience, not a partial translation layered onto an English-first product	The platform interface, including full right-to-left (RTL) layout, is available in Arabic and English as a core, included capability
Compatibility with the organisation's internal cybersecurity review and approval process before go-live	The platform is developed with awareness of the OWASP Top 10 and includes controls such as role-based access, audit logging, and API rate limiting; the full architecture is made available for client security review, and deployment is subject to the client's cybersecurity assessment as described in Section 19
Practical integration paths into the organisation's existing systems (identity, HR, reporting) rather than a fully closed platform	REST APIs and a defined set of integration options are available; specific integrations such as single sign-on, SAP SuccessFactors (or other HR systems), or Power BI can be configured to support PDO 's environment and are scoped individually, as described in Section 22

Local support that understands the organisation's language, working hours, and escalation expectations	Altanfith is based in the Sultanate of Oman and delivers support in Arabic and English under Bronze, Silver, and Gold packages, as described in Section 30
--	--

These requirements shape both the design of the platform and the structure of this proposal: the sections that follow set out how the Live Quiz Platform, its architecture, and our delivery methodology are built to answer each of them directly.

8. Business Challenges

Organisations that rely on free or general-purpose public quiz tools for internal engagement, training, or assessment activities commonly encounter a consistent set of limitations once usage moves beyond a single informal session.

Challenge	Consequence
Generic public quiz tools are built for open, consumer use and lack enterprise controls	No organisational branding, no role-based administration, and no ability to enforce the organisation's own access and data-handling requirements
Quiz and question content is created ad hoc and scattered across individual accounts, files, and departments	Content cannot be reused, standardised, or governed centrally, and is often lost when the individual who created it moves on
Public tools offer little to no organisational-level reporting	Management cannot see participation, performance, or engagement trends across departments, sessions, or time — results exist only as isolated, per-session snapshots
Data created and collected through a public, third-party tool sits outside the organisation's own environment and control	This raises data residency and governance concerns for organisations subject to internal compliance requirements or operating in regulated sectors, particularly where session content or participant data is sensitive
Repetitive use of the same generic tool and format across HSE briefings and training sessions leads to engagement fatigue	Participation and attention decline over time, undermining the safety and learning objectives the sessions are intended to support
Public tools typically offer only a single, English-first interface	Arabic-speaking participants and administrators are left with a translated or inconsistent experience, reducing usability and adoption across a bilingual workforce

The Live Quiz Platform, and the enterprise delivery model set out in this proposal, are designed specifically to address these gaps — as described in the sections that follow.

PART TWO

How the Live Quiz Platform answers the requirements and challenges set out above

9. Proposed Solution

Altanfith proposes the Live Quiz Platform as a single platform that brings together live engagement, structured assessment, and organisational analytics, so that **PDO** does not need to operate separate tools for running a live session, evaluating knowledge, and reporting on the results.

The platform is designed to support a broad range of use cases across a large organisation, from safety and training activities to recruitment and internal communication. The table below reflects the categories of use the platform is built to support; the specific mix relevant to **PDO** can be confirmed during discovery.

Engagement & Training	Assessment & Evaluation	Organisational & Institutional
HSE meetings	Knowledge assessments	Government awareness campaigns
Employee training	Recruitment assessments	Educational institutions
Corporate workshops	Interviews	Schools & universities
Team-building	Surveys & polls	Internal compliance assessments
Corporate events	Certification preparation	Employee onboarding
Competitions	Learning & development programmes	Safety awareness

It is important to be clear about how we position the Live Quiz Platform. Live, buzzer-style quiz tools such as Kahoot are widely known and are a useful category reference for the live-session experience — but the Live Quiz Platform is not a copy of any such tool. It is an enterprise platform: customisable to **PDO**'s operational workflows, branding, reporting requirements, security posture, and integration landscape, and deployable in a way a general public tool is not, including fully on-premises within the client's own infrastructure. The sections that follow set out the platform itself (Sections 10–15), its technical architecture and deployment options (Sections 16–23), our delivery methodology (Sections 24–31), and the commercial terms under which it is offered (Sections 32 onward).

10. Platform Overview

The Live Quiz Platform is best understood through the lifecycle of a typical session, from creation through to reporting.

An administrator or quiz creator begins by building a quiz manually, or by using the platform's AI-assisted generation to produce a complete quiz from a topic, an uploaded PDF or Word document, a URL, or pasted text — with every AI-generated question reviewed and editable before use. The quiz is then branded with the organisation's own logo, colours, and theme, and can be organised into a reusable question bank tagged by subject and difficulty for future sessions.

When it is time to run a session, the host launches the quiz live. Participants join in seconds using a QR code, a direct link, or a session PIN, without needing to create an account. As the session runs, the platform is server-authoritative for timing and scoring: the server determines when a question opens and closes and calculates results, while participant and host devices render what is happening in real time — including live answer submission, a live leaderboard, and automatic reconnection if a participant's network connection drops mid-session. At the close of the session, the top three participants are shown on a podium, alongside the full participant ranking.

After the session ends, results are not lost. They are captured in the platform's reporting and session history, giving administrators and management access to participation data, performance analysis, and historical comparisons across sessions — described in full in Section 15.

Across this lifecycle, the platform's capability set is organised into six pillars, summarised below and detailed in full in Section 12.

Pillar	What it covers
Quiz creation & content	Quiz and question authoring, question banks, bulk import, templates, versioning, and auto-save/draft recovery
Live hosting & participation	QR/link/PIN joining, real-time answers, live leaderboard, ranking, podium, and automatic reconnection
Branding & experience	Organisation branding, custom themes and backgrounds, and a bilingual Arabic/English interface with full RTL support
Reporting & analytics	Result summaries, performance and difficulty analysis, historical and department-level reporting, and Excel/CSV/PDF export
Administration & governance	Role-based access, organisation and department structure, multiple administrators, audit logs, and session history
AI capabilities	AI-assisted quiz generation from topic, document, URL, or text, with administrator review before use — detailed in Section 13

11. Current Use Case

The Live Quiz Platform is not a concept, a prototype, or a proof of concept — it is in real, ongoing operational use today.

The platform is currently deployed and actively used within **Abraj Energy Services**, where it supports HSE (Health, Safety, and Environment) meetings and employee engagement activities. Live sessions run on the platform bring participants together to join and answer in real time within a single session.

We present this use case as evidence that the core platform — live session hosting, real-time participation at meaningful scale, and the underlying engagement experience — is production-proven and functioning in a real organisational setting, rather than as a marketing claim or a controlled

demonstration environment. We have not included any additional client references, dates, or testimonial statements beyond what is described above. All references in this proposal are drawn from this production deployment.

12. Functional Scope

The tables below set out the platform's functional scope in full, grouped into six capability areas. Each capability is marked with its current status: **Included** capabilities are part of the core platform today; **Optional** capabilities are available as scoped additions; **Future enhancement** items are on our roadmap and not part of the current platform.

12.1 Quiz Creation & Content

Capability	Description	Status
Quiz creation & management	Create, edit, and manage quizzes from a central administration interface	Included
Multiple question types	Single-answer questions with 4–6 choices, multi-select questions, and true/false questions	Included
Polls & surveys	Non-scored question formats for gathering opinion or feedback	Included
Question timers	Configurable time limits per question	Included
Scoring & points configuration	Configurable points and scoring rules per question or quiz	Included
Public & private quizzes	Quizzes can be restricted to the organisation or made available more broadly, per administrator configuration	Included
Question bank	Central, reusable question repository organised by subject, tags, and difficulty	Included
Bulk import	Bulk import of questions from Excel, CSV, or Word files	Included
Quiz templates & reuse	Reusable quiz templates to standardise content across sessions and departments	Included

Edit & duplicate quizzes	Existing quizzes can be edited or duplicated as a starting point for new sessions	Included
Auto-save during creation	Quiz content is saved automatically while an administrator is authoring it	Included
Draft recovery	In-progress drafts can be recovered rather than lost	Included
Versioning of quiz content	Changes to quiz content are versioned over time	Included

12.2 Live Hosting & Participation

Capability	Description	Status
Live quiz hosting	Host a quiz as a live, timed session for a group of participants	Included
QR-code joining	Participants join a session by scanning a QR code	Included
Direct-link joining	Participants join via a direct shareable link	Included
Game code / session PIN	Participants join by entering a short session PIN	Included
Real-time participant joining	Participants appear in the host's lobby as they join, in real time	Included
Real-time answer submission	Participant answers are submitted and processed in real time via WebSocket connection	Included
Real-time leaderboard	A live leaderboard updates as answers are scored	Included
Live ranking	Participant ranking updates live throughout the session	Included
Top-three winners podium	A podium view highlights the top three participants at session close	Included
Full participant ranking	Complete ranking of all participants, not only the top three	Included

Automatic reconnection	Participants who lose network connectivity mid-session are automatically reconnected	Included
------------------------	--	----------

12.3 Branding & Experience

Capability	Description	Status
Custom themes	The platform visual theme can be configured to match the institution's branding	Included
Custom backgrounds	Custom background imagery for quizzes and sessions	Included
Organisation branding	Organisation branding, including logo and colour customisation	Included
Custom UI text	Interface text can be adapted to the institution	Included
Arabic & English interface with full RTL	Complete bilingual interface with full right-to-left layout support for Arabic — positioned as a core enterprise bilingual capability, not an add-on translation layer	Included
Additional languages	Support for interface languages beyond Arabic and English	Optional

12.4 Reporting & Analytics

Reporting and analytics are part of the platform's functional scope. The full capability set — result summaries, participant rankings, answer and question-difficulty analysis, participant completion, department-level and historical reporting, session comparison, export to PDF, Excel, and CSV with Arabic and English report headers, the management dashboard, and AI-generated insights (future enhancement) — is set out in **Section 15 (Reporting and Analytics)**.

12.5 Administration & Governance

Capability	Description	Status
Administrative dashboard	Central dashboard for platform and content administration	Included
Organisation user accounts	User accounts scoped to the organisation	Included

Role-based access	Access and permissions governed by role-based access control	Included
Multiple departments	Support for multiple departments within the organisation, each with its own isolated data	Included
Multiple administrators	Support for more than one administrator within the organisation	Included
Audit logs of administrative actions	Administrative actions are recorded in an audit log	Included
User activity/login visibility	Visibility into user activity and login events	Included
Quiz session history	Historical record of quiz sessions run	Included
Soft-delete with archive	Deleted content is archived rather than permanently removed, supporting recovery	Included
Notification options	Configurable notifications for administrators and/or participants	Optional

12.6 AI Capabilities

Capability	Description	Status
AI-assisted content generation	Generate quizzes from a topic, an uploaded PDF or Word document, a URL, or pasted text, with administrator review before use. The full classification of current, optional, and future AI capabilities is set out in Section 13 (AI Capabilities)	See Section 13

13. AI Capabilities

The Live Quiz Platform includes AI-assisted content generation as a core, already-implemented capability, alongside a further set of AI features that are available as optional, separately scoped additions, and a set of AI-driven reporting capabilities that sit on our future roadmap. This section sets out the distinction clearly, so that **PDO** can see exactly what is included today, what can be added, and what is planned for the future.

13.1 Current Capabilities (Included)

Capability	Description	Status
Quiz generation from a topic or title	A complete quiz, including questions and answer choices, is generated from a short topic or title prompt	Included
Quiz generation from an uploaded PDF	Questions are generated directly from the content of an uploaded PDF document	Included
Quiz generation from an uploaded document	Questions are generated directly from the content of an uploaded Word document	Included
Quiz generation from a URL	Questions are generated from the content of a supplied URL or webpage	Included
Quiz generation from pasted text	Questions are generated directly from text pasted into the platform	Included
AI-assigned difficulty levels	Generated questions are automatically assigned a suggested difficulty level	Included
AI-generated answer explanations	Generated questions include a suggested explanation for the correct answer	Included
Administrator review before use	All AI-generated content is presented to an administrator for review and editing before it is used in a live session or published to a question bank	Included

13.2 Optional Capabilities (Scoped per Engagement)

The following AI capabilities are not part of the current platform but can be added as a separately scoped engagement.

Capability	Description	Status
AI question suggestions	AI-suggested questions offered while an administrator authors a quiz manually	Optional
AI answer suggestions	AI-suggested answer choices for a manually authored question	Optional
AI question rewriting	AI-assisted rewriting of existing question text for clarity or tone	Optional

AI-assisted difficulty re-adjustment	AI-assisted review and re-adjustment of difficulty levels across an existing question bank	Optional
AI-generated quiz backgrounds	AI-generated background imagery for quizzes and sessions	Optional
AI-generated themes	AI-generated visual themes for the platform interface	Optional
AI-assisted result summaries	AI-assisted narrative summary of a single session's results	Optional

13.3 Future Roadmap

Capability	Description	Status
AI-assisted insights over historical results	AI-generated narrative insights and trend analysis across historical session data	Future enhancement
AI-generated management reports	AI-generated summary reports intended for management-level review	Future enhancement

AI features are delivered using a commercial large-language-model API; usage allowances, and any associated API costs, are defined in the applicable commercial model (see Section 32). Cloud-based AI generation requires internet connectivity. A private, on-premises AI deployment is a future roadmap option that would depend on separate, dedicated infrastructure and is not part of the current platform. As with any AI-generated content, **PDO** should review AI-generated quizzes, questions, and summaries before they are used in a live or published context.

AI and data residency. AI-assisted content generation may utilise approved cloud-based large-language-model (LLM) services. Only the minimum information required to generate the requested content is transmitted, and participant personal data is not required for AI content generation. AI capabilities can be configured, restricted, or disabled according to **PDO**'s information security, privacy, and data-residency requirements, and a private AI deployment is available as a future deployment option on the platform roadmap (Section 42) rather than as part of the current platform.

14. User Roles

The platform's role-based access model is designed to reflect how a large organisation actually assigns administrative and operational responsibility, rather than offering a single flat administrator account.

Role	Purpose
------	---------

Super Administrator	Full platform administration across all organisations and departments
Organisation Administrator	Manages a single organisation, including its users, branding, and content
Department Administrator	Manages a department's content and users within an organisation
Quiz Creator	Authors quizzes and question banks
Trainer	Runs training-oriented sessions
Host	Runs live sessions
Viewer	Views results
Participant	Joins and answers in a live session — no account required
Auditor	Read-only access to audit and activity logs
Read-Only User	Read-only access to dashboards and reports

The final role model applied for **PDO** is configured during discovery, and can be mapped to the client's own directory groups where a single sign-on integration is included in scope (see Section 22).

Licensing for the Live Quiz Platform places no numerical limit on named administrative accounts: **PDO** may provision an unlimited number of Super Administrators, Organisation Administrators, Department Administrators, Hosts, Trainers, and Quiz Creators. Participant access is likewise unlimited: participants join live sessions via session PIN, QR code, or direct link, without a named account. No licensing cap is applied to administrators or participants; concurrent live-session capacity is a separate, technical measure, governed by the performance considerations set out in Section 23.

15. Reporting and Analytics

The platform's reporting and analytics capabilities give administrators and management visibility into participation, performance, and engagement, both at the level of a single session and across the organisation's session history over time.

Capability	Description	Status
Quiz result summary	Summary results at quiz and session level	Included

Participant ranking	Ranked results per participant	Included
Top performers	Identification of top-performing participants	Included
Correct/incorrect answer analysis	Breakdown of correct versus incorrect responses	Included
Question difficulty analysis	Analysis of question-level difficulty based on response patterns	Included
Participant completion	Visibility into who completed a session and who did not	Included
Department-level reporting	Reporting rolled up by department or organisational unit	Included
Historical reports	Access to results and reports from past sessions	Included
Session comparison	Comparison of results across multiple sessions	Included
Export to PDF	Report export in PDF format	Included
Export to Excel & CSV	Report export in Excel and CSV formats	Included
Management dashboard	Consolidated dashboard view for management-level oversight	Included
AI-generated insights	AI-generated narrative insights over historical results	Future enhancement

All PDF, Excel, and CSV exports are compliance-ready: report headers are rendered in both Arabic and English, so that reports generated for **PDO** can be shared and archived internally without additional reformatting, regardless of which language the reviewing team works in.

PART THREE

How the platform is built, deployed, and secured

16. Technical Architecture

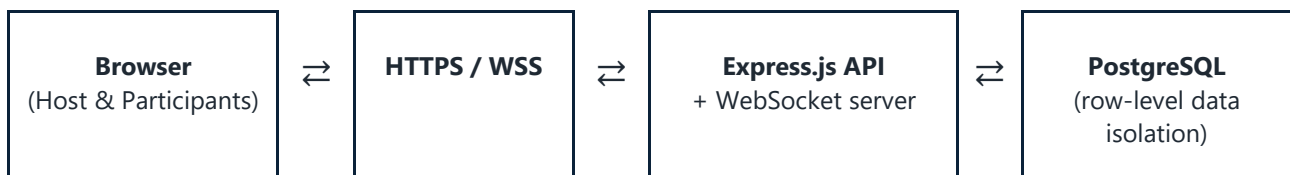
The Live Quiz Platform is built on a modern, well-established web application stack, selected for reliability, security, and the ability to run in cloud, on-premises, or hybrid environments, subject to the infrastructure assessment described in Section 18.2.

16.1 Technology Stack

Layer	Technology	Notes
Frontend	React (single-page application)	Also deployable with Next.js where required by the client's environment
Backend	Node.js with Express.js	Exposes a REST API consumed by the frontend
Authentication	Token/session-based authentication	JWT or an equivalent secure token-based system
Database	PostgreSQL (recommended)	Enterprise database with row-level data isolation; MongoDB may be considered depending on the final architecture agreed with PDO
Real-time layer	WebSockets	Used for participant joining, live answers, and leaderboard updates; timing and scoring are server-authoritative
Hosting	Cloud, on-premises, or hybrid	See Section 18 for the full comparison of deployment options

16.2 Architecture Diagram

The diagram below shows the platform's core request and real-time data flow, followed by the three supported deployment variants.



Browser clients communicate with the Express.js API and WebSocket server over HTTPS and secure WebSockets (WSS); the application server is the sole component with direct database access.

Deployment Variant	Application Layer (Frontend + Express.js API)	Database Layer
--------------------	---	----------------

Cloud / SaaS	Altanfith Managed Cloud Infrastructure	Altanfith Managed Cloud Infrastructure
On-Premises	Client enterprise infrastructure environment	Client enterprise infrastructure environment
Hybrid	Altanfith Managed Cloud Infrastructure	Client enterprise infrastructure environment (sensitive data remains client-side)

17. Real-Time Architecture

Live sessions depend on the platform's real-time layer behaving predictably under load, and behaving fairly for every participant regardless of network conditions. This section describes how that is achieved.

17.1 Session Flow

A live session follows a consistent, server-driven flow from join through to the final leaderboard.

Step	What happens
1. Join	Participants connect over a WebSocket session as they scan a QR code, follow a direct link, or enter a session PIN; the host's lobby view updates in real time as each participant joins
2. Question broadcast	When the host opens a question, the server broadcasts the question to all connected participants at the same moment
3. Answer submission	Participants submit answers over the WebSocket connection while the question is live; the correct answer is never sent to a participant's device before the question closes
4. Server-side scoring	The server, not the client, determines when the question closes, evaluates each submitted answer, and calculates the resulting score
5. Leaderboard push	Updated rankings are pushed to the host and participant devices in real time once scoring is complete

17.2 Server-Authoritative Timing, Reconnection, and Why It Matters at Scale

Question timing, closure, and scoring are controlled by the server, not by any individual device: participant and host devices render a countdown, an open or closed question, and an updated leaderboard, but they never decide when a question opens or closes and are never trusted with the correct answer before it is safe to reveal it. Where a participant's device loses connectivity mid-session

— common in venues with variable Wi-Fi or mobile coverage — the platform automatically attempts to reconnect the session rather than dropping the participant. For a session of the scale described in Section 23, this combination is what keeps a large, live audience in a fair, synchronised session rather than a collection of individually drifting client experiences.

18. Deployment Options

PDO can choose from three deployment options, so that the choice of hosting model reflects data residency, network, and cybersecurity requirements rather than a one-size-fits-all default.

	Option 1 — SaaS / Cloud	Option 2 — On-Premises	Option 3 — Hybrid
Hosted by	Altanfith	Client infrastructure	Split between Altanfith and client infrastructure
Includes	Cloud hosting, server & database management, backups, security updates, monitoring, maintenance, technical support, version upgrades, AI usage allowance	Application deployment, backend installation, database configuration, environment setup, initial cybersecurity configuration, documentation, training, knowledge transfer	Application components hosted by Altanfith; sensitive data/database remain in the client environment
Commercials	Annual subscription; lower initial investment; fastest implementation	Project fee; optional source-code handover; optional annual maintenance contract	Per final design
Client provides	Browser access only	Hardware, OS, internal networking, data-centre requirements (unless in scope)	Per final design

18.1 Cloud / SaaS

Altanfith hosts and manages the full application and database layer on Altanfith Managed Cloud Infrastructure, so **PDO** requires only browser access. Hosting, backups, monitoring, security updates, and version upgrades are managed by Altanfith as part of the subscription (Section 32, Model A).

18.2 On-Premises

The Express.js backend runs on the client's own internal servers or private cloud — a public cloud environment is not required. The platform's backend has been built to run within a range of internal environments, including:

Supported Environment

Client data centre	Internal servers
Private cloud	Virtual machines
Docker containers	Kubernetes
Linux servers	Windows Server (subject to technical validation)
Hybrid combinations of the above infrastructure types	

On-premises deployment is **subject to infrastructure assessment**. Network policies, database selection, internal DNS, SSL certificates, firewall configuration, backup procedures, and **PDO's** cybersecurity requirements are all confirmed during this assessment before deployment proceeds. Altanfith does not promise universal compatibility with every client environment in advance of this assessment.

A typical on-premises deployment draws on the following client-provided components: a virtual machine or container platform; a Linux server environment; a PostgreSQL database; a reverse proxy; TLS certificates; internal DNS; firewall rules; and outbound HTTPS connectivity where cloud-based AI features are in scope. Minimum infrastructure sizing will be confirmed during the Technical Assessment phase (Section 24, Phase 2).

18.3 Hybrid

A hybrid deployment splits the platform between Altanfith-hosted application components and client-hosted, sensitive data storage. The final design of a hybrid deployment depends on **PDO's** specific security and integration requirements, and is agreed during discovery and infrastructure assessment rather than fixed in advance.

18.4 Supported Deployment Platforms

Under the SaaS and hybrid options, the platform is operated on **Altanfith Managed Cloud Infrastructure**. The underlying deployment platform for each engagement is selected according to the client's IT strategy, security requirements, and data-residency requirements, from the following supported platforms:

Supported Platform	Typical Use
Microsoft Azure	Organisations standardised on Microsoft cloud services
Amazon Web Services (AWS)	Organisations standardised on AWS
Google Cloud Platform (GCP)	Organisations standardised on Google cloud services

Oman Data Park	In-country hosting where data residency within the Sultanate of Oman is required
Private Cloud	Client-selected private cloud environments
Client On-Premises Infrastructure	Fully client-hosted deployment, as described in Section 18.2

The deployment platform selected for **PDO** is confirmed during discovery and infrastructure assessment (Section 24, Phases 1–2).

18.5 Client Standard Requirements

The final solution will be designed, configured, and deployed in accordance with **PDO**'s approved enterprise standards and technical requirements identified during the Discovery and Technical Assessment phases (Section 24, Phases 1–2). The following are confirmed with **PDO** before implementation:

Infrastructure Standards	Hosting Standards
Network Requirements	Cybersecurity Requirements
Identity & Access Management Standards	Integration Standards
API Standards	Data Residency Requirements
Data Classification Requirements	Backup & Disaster Recovery Requirements
Monitoring & Logging Standards	Compliance Requirements
Business Continuity Requirements	High Availability Requirements
Performance Requirements	Operational Support Requirements

These requirements apply regardless of the deployment model selected — Altanfith Managed Cloud Infrastructure, client on-premises infrastructure, or a hybrid deployment — and implementation always aligns with **PDO**'s approved IT policies, security standards, governance framework, and operational procedures.

19. Cybersecurity

The Live Quiz Platform is designed with cybersecurity as a foundational concern rather than an add-on, and its controls are developed with awareness of the OWASP Top 10. At the same time, we are deliberate about how we describe our security posture: the statements in this section describe how the platform is designed and configured, not a formal certification, and final acceptance of the security architecture remains subject to **PDO**'s own cybersecurity assessment.

Domain	Controls
1. Encryption & transport	HTTPS/TLS; encryption in transit; encryption at rest where applicable; backup encryption
2. Identity & access	Secure (bcrypt) password hashing; role-based access control (RBAC); least-privilege access; strong password policy; multi-factor authentication (Optional); session management; token expiration; secure API authentication; user account deactivation; regular access reviews; role-validated host-only operations
3. Application security	API rate limiting; input validation; output encoding; protection against common web vulnerabilities; secure development practices; developed with awareness of the OWASP Top 10; secure API configuration
4. Audit & monitoring	Audit logs of administrative actions; login logs; failed-login monitoring; data-access logging; monitoring & alerting; log retention (audit and security logs retained for 12 months by default; see Section 20)
5. Data lifecycle & resilience	Backup policy; backup retention; disaster recovery planning; business continuity planning; data retention rules; data deletion and export procedures — expanded in Section 20
6. Infrastructure & operations	Secure environment configuration; firewall configuration; network segmentation (on-premises); row-level data isolation in PostgreSQL; database access restrictions; secrets management; environment-variable management; secure source-code repository; separated dev/test/prod environments
7. Process	Change management; incident response; security incident notification; data-breach escalation; secure release process; security patching; vulnerability assessment; penetration testing (Optional service); client-approved security architecture

These controls are designed to align with recognised good practice and with awareness of the OWASP Top 10 — not represented as certified or compliant. Final acceptance is confirmed through **PDO**'s own security assessment (Phase 3, Section 24); the security architecture is only finalised once client-approved. Formal penetration testing is available as an **optional service**, quoted separately (Section 32); it is not included in the base engagement.

Compliance alignment. The platform is designed to align with the Oman Personal Data Protection Law (Royal Decree 6/2022) and can be configured to support **PDO**'s information security requirements. Its

controls are designed to align with ISO/IEC 27001 control themes, the OWASP Application Security Verification Standard (ASVS), and the NIST Cybersecurity Framework, where applicable. These statements describe design intent and configurability rather than certification or formal accreditation, and remain subject to **PDO**'s own security assessment (Section 24, Phase 3).

20. Data Protection

Data protection builds on the data lifecycle and resilience controls introduced in Section 19 (Domain 5), and describes, in more detail, how **PDO**'s data is backed up, retained, deleted, and exported. Availability and data-durability commitments are not presented as absolutes; they are defined in the agreed SLA, and the platform is designed to align with sound backup, retention, and recovery practice within the parameters agreed with **PDO**.

Aspect	Approach
Backups	Operated under a defined backup policy; frequency, encryption, and retention arrangements are confirmed as part of the applicable deployment option (Section 18) and the commercial agreement
Retention	Session results and participant records are retained for 24 months , and audit and security logs for 12 months , by default. Data retention policies are configurable according to PDO 's governance, regulatory, and business requirements, subject to confirmation during discovery
Deletion	Follows a defined procedure: soft-delete with archive is available at the content level (Section 12.5); permanent deletion procedures, including at contract termination, are defined in the commercial agreement
Export	Available on request, and provided as a standard step at contract termination, in an agreed format, so PDO is not dependent on continued platform access to retain its own data
Disaster recovery & business continuity	Addressed as part of the deployment design and configured to support PDO 's requirements; specific recovery time/point objectives are confirmed during discovery and infrastructure assessment and depend on the deployment option selected

21. Data Ownership

Ownership of data and intellectual property is set out clearly, so that **PDO** understands exactly what it owns, what Altanfith owns, and how this is affected by the commercial model selected.

- Client data remains the property of **PDO** at all times.
- Altanfith claims no ownership of client operational data.
- Data export is provided at contract termination, in an agreed format.

- Retention and deletion terms are defined in the commercial agreement.
- For on-premises deployments, data may remain entirely within **PDO**'s own enterprise infrastructure environment.
- Intellectual property ownership depends on the commercial model selected (see Section 32).
- Standard platform IP remains with Altanfith unless transferred by separate agreement.
- Client-specific content — including questions, participant data, and reports — remains the property of **PDO**.
- Source-code ownership is not automatically included, and applies only where it is expressly stated in the commercial agreement.

22. Integration Options

The Live Quiz Platform exposes REST APIs that support integration into **PDO**'s existing systems. Integrations are delivered as scoped work under the Enterprise Custom Development Partnership commercial model (Model C, see Section 32), rather than assumed as part of the base platform, and their final scope is always subject to the availability and technical constraints of the relevant third-party API.

Integration	Description	Status
Microsoft Entra ID (Azure AD) single sign-on	Authentication against the client's existing identity provider, with roles mapped to directory groups	Optional
Microsoft Active Directory	Directory-based authentication and role mapping for environments not yet migrated to Microsoft Entra ID	Optional
Microsoft Teams	Integration supporting session distribution or notification via Microsoft Teams	Optional
Microsoft Power BI	Integration exposing platform data for consumption in Power BI dashboards	Optional
SAP SuccessFactors	Integration with the client's HR platform, for example for user or organisational-unit synchronisation	Optional
Learning Management System (LMS)	Integration with the client's LMS for training-related workflows	Optional
REST APIs	Documented platform APIs supporting client-built integrations and data extraction	Included
SCIM provisioning	Automated user provisioning and de-provisioning from the client's identity platform	Optional

CSV import/export	Bulk import and export of users, questions, and results in CSV format	Included
Additional client systems	Any additional client system identified during discovery	Optional

Except for the platform's documented REST APIs and CSV import/export, which are standard platform capabilities, the integrations listed above are optional and subject to the availability, documentation, and technical constraints of the relevant third-party API; final integration scope is confirmed during discovery (see Section 24, Phase 1) and delivered under Phase 8 (Integration Development) of the implementation methodology. Integrations are implemented according to the client's environment and the availability of the relevant APIs.

23. Performance and Scalability

Performance and scalability claims for the Live Quiz Platform are grounded in what has actually been observed, rather than presented as an unqualified guarantee.

In real production use at Abraj Energy Services (Section 11), the platform supports live organisational sessions. It has additionally been validated through internal load testing for **400+ concurrent participants per live session** against the production configuration, giving **PDO** a tested reference point beyond day-to-day operational use. Higher capacities can be supported following infrastructure sizing, deployment optimisation, and dedicated performance testing.

Beyond these two reference points, the following considerations apply to any deployment for **PDO**:

- Final concurrency targets are confirmed during discovery, against the validated baseline of **400+ concurrent participants per live session**.
- Load testing is recommended ahead of any enterprise deployment with a large expected audience.
- Cloud scaling and infrastructure sizing are based on the expected usage profile agreed with **PDO**.
- For on-premises deployments, real-time performance depends on the client's own infrastructure and network configuration.
- Real-time functionality requires proper network and server configuration to perform as designed.
- Performance monitoring may be included as part of the applicable support and maintenance package (see Section 30).

Scaling beyond the levels referenced above is planned and validated against concurrency targets confirmed with **PDO** during discovery, rather than presented as an open-ended capability. Performance at higher concurrency levels than those referenced above should be validated through load testing scoped to **PDO**'s specific requirements before a large-scale go-live.

PART FOUR

How Altanfith plans, governs, and delivers the engagement

24. Implementation Methodology

Altanfith delivers the Live Quiz Platform through a structured, 17-phase methodology, designed to take **PDO** from initial requirements through to a supported, live production system with clear exit criteria at each stage.

Before implementation begins, **Altanfith** and **PDO** jointly validate the foundations on which delivery depends: infrastructure readiness, security controls, integration architecture, network connectivity, identity management, compliance obligations, data governance, performance expectations, and operational acceptance criteria. This validation is carried out through the Discovery, Technical Assessment, and Cybersecurity Review phases (Phases 1–3) and is reflected in their exit criteria, so that implementation proceeds on jointly confirmed facts rather than assumptions.

Phase	Purpose & Key Activities	Exit Criteria
1. Discovery & Requirements Gathering	Structured workshops and stakeholder interviews establish functional, branding, and organisational requirements	Documented and agreed requirements
2. Technical Assessment	Reviews the client's enterprise infrastructure, network, and hosting environment	Assessment report; confirmed deployment option
3. Cybersecurity Review	Engages the client's security team on the proposed architecture and controls (Section 19)	Client-approved security architecture
4. Solution Design	Translates requirements into a detailed solution design, incl. configuration, branding, and custom/integration scope	Approved solution design document
5. UI & Branding Customisation	Applies the client's logo, colour palette, themes, and UI text	Branded environment reviewed by client
6. Configuration	Configures organisations, departments, user roles, and platform settings	Configured environment ready for build/test
7. Custom Development	Builds agreed custom features, workflows, or reports	Custom development complete and reviewed
8. Integration Development	Builds and configures agreed integrations (Section 22)	Integrations functioning against client systems

9. Internal Testing	Functional and regression testing across the configured environment	Test results reviewed; defects triaged
10. Security Testing	Security-focused testing against the Section 19 controls	Testing complete; remediation applied
11. User Acceptance Testing	Client tests the platform against agreed acceptance criteria (Section 29)	UAT sign-off (Milestone M5)
12. Data Migration (if required)	Migrates agreed legacy data into the platform, where in scope	Migrated data validated against source
13. Deployment	Deploys the platform to the production environment	Production environment live and verified
14. Training	Administrator, quiz-creator, and support-team training (Section 28)	Training completed; materials handed over
15. Go-Live	Platform goes live for production use, following successful UAT	Go-live checklist completed (Milestone M6)
16. Hypercare	Heightened, close-support monitoring immediately following go-live	Hypercare completed; no unresolved critical issues
17. Transition to Support	Hands the platform from the project team to ongoing support (Section 30)	Support handover complete (Milestone M7)

25. Project Timeline

The table below presents a sample timeline of 6–12 weeks, mapped against the phases described in Section 24.

Weeks	Phases
W1–2	Discovery & Requirements Gathering; Technical Assessment; Cybersecurity Review
W2–4	Solution Design; UI & Branding Customisation

W3–7	Configuration; Custom Development; Integration Development
W7–9	Internal Testing; Security Testing; User Acceptance Testing
W9–10	Deployment; Training
W10	Go-Live
W10–12	Hypercare; Transition to Support

25.1 Milestones

Milestone	Description	Indicative Completion	Sign-off by
M1	Requirements sign-off	W2	Client Project Manager
M2	Solution design approval	W4	Client Project Manager
M3	Configured & branded environment ready	W7	Altanfith Project Manager
M4	Integrations complete	W7	Altanfith Technical Lead
M5	UAT sign-off	W9	Client Project Manager
M6	Go-live	W10	Project Sponsor
M7	Hypercare exit / support handover	W12	Client Project Manager & Altanfith Project Manager

This timeline is indicative only. The actual project timeline for **PDO** depends on the final agreed scope, the number and complexity of integrations, and the timeliness of client approvals at each milestone.

26. Project Governance

The engagement is governed through a clearly defined set of roles and a regular reporting cadence, so that progress, issues, and decisions are visible to both parties throughout delivery.

Role	Responsibility
Project Sponsor (client)	Owns the business case and provides executive-level decisions and escalation support
Client Project Manager	Coordinates client-side stakeholders, approvals, and resourcing
Altanfith Project Manager	Coordinates delivery, reporting, and issue/risk management on behalf of Altanfith
Technical Lead (Altanfith)	Owns technical design, development, and integration decisions
Cybersecurity Representative (client)	Reviews and approves the security architecture and controls
Business Owner (client)	Confirms functional requirements and acceptance criteria on behalf of the business

Governance is supported by a consistent set of cadence and mechanics: weekly status meetings; regular progress reporting; a maintained issue log and risk log; a formal change-request management process for scope changes; an escalation process running from project manager to project sponsors; and a documented sign-off process at each milestone described in Section 25.

27. Deliverables

The table below lists the deliverables produced over the course of the engagement.

Deliverable	Description	Included?
Requirements document	Documented, agreed functional and non-functional requirements	✓
Solution design document	Detailed design of the configured and/or customised solution	✓
Configured platform	The platform configured per the agreed solution design	✓
Customised UI	Branding and UI customisation applied per Section 24, Phase 5	✓

Deployed frontend	The React frontend deployed to the agreed environment	✓
Deployed Express backend	The Express.js API and WebSocket server deployed to the agreed environment	✓
Database setup	Database configured and provisioned per the solution design	✓
Security configuration	Security controls configured per the client-approved security architecture	✓
User roles configured	Role-based access configured per the agreed role model	✓
Administrator account setup	Initial administrator accounts provisioned	✓
Reports	Reporting and analytics configured per Section 15	✓
Training material	Materials used in the training described in Section 28	✓
User manual	End-user reference documentation	✓
Administrator manual	Administrator reference documentation	✓
Deployment guide	Documentation of the deployed environment	✓
Backup guide	Documentation of backup procedures for the deployed environment	✓
Support handover	Formal handover to the ongoing support team (Phase 17)	✓
UAT plan	Agreed User Acceptance Testing plan and criteria (Section 29)	✓
Test report	Summary of internal, security, and acceptance testing outcomes	✓

Go-live checklist	Checklist confirming readiness for production go-live	✓
Source code	Provided only where included in the selected commercial model	Only if included in the commercial model
API documentation	Provided where included in scope	If included

28. Training and Knowledge Transfer

Training is provided so that **PDO**'s own team can operate, administer, and support the platform confidently after go-live.

Training Item	Description	Status
Administrator training	Training for administrators on platform configuration and management	Included
Quiz-creator training	Training for staff authoring quizzes and question banks	Included
Support-team training	Training for the client's internal support or IT team	Included
User manuals	Reference documentation for end users	Included
Quick-start guide	Condensed getting-started reference	Included
Recorded training	Recorded training sessions for later reference or onboarding new staff	Optional
Train-the-trainer	Enables client-side trainers to deliver ongoing internal training	Optional
Arabic and English training material	Bilingual training material	Optional

29. Testing and Acceptance

The platform passes through three layers of testing before it is accepted for production go-live. Altanfith performs internal functional and regression testing (Phase 9) across the configured

environment, followed by security-focused testing (Phase 10) against the controls described in Section 19, ahead of client security review and sign-off. User Acceptance Testing (Phase 11) is then carried out against a client-approved UAT plan and agreed acceptance criteria, so that **PDO** — not Altanfith alone — confirms that the configured platform meets its requirements before go-live.

Any defects identified during internal, security, or acceptance testing are triaged using the same four-level severity model used for support (Section 31): Critical, High, Medium, and Low. UAT sign-off is Milestone M5 in the project timeline (Section 25), and it gates go-live: production rollout depends on successful UAT.

30. Support and Maintenance

Ongoing support is delivered under one of three packages, so that **PDO** can select the level of coverage and responsiveness that matches its operational needs.

Package	Includes
Bronze	Oman business-hours support; email support; bug fixes; standard updates; best-effort response
Silver	Everything in Bronze, plus: priority support; faster response times; monthly system health review; performance optimisation; minor improvements; remote assistance
Gold	Everything in Silver, plus: critical incident support; dedicated support contact; priority issue resolution; proactive monitoring; security monitoring; quarterly service review; emergency support; 24/7 support (Optional); dedicated development allocation (Optional)

31. SLA

Service levels are defined by severity, with response time and target resolution time tracked as two separate measures — response confirms that an issue has been received and is being actioned; resolution is the target time to resolve it, and is not a guarantee.

Severity	Definition	Response Time	Target Resolution Time
1 — Critical	Platform down / live session blocked for all users	1 business hour	Workaround within 4 business hours; permanent fix within 2 business days
2 — High	Major function degraded, no workaround	4 business hours	2 business days

3 — Medium	Limited impact, workaround exists	1 business day	5 business days
4 — Low	Cosmetic / question / minor request	2 business days	Next planned release

Response and target resolution times are separate measures: response time confirms an issue has been received and triaged; target resolution time is a goal, dependent on the agreed SLA, and is not a guaranteed outcome. Actual values depend on the support package selected (Section 30) and the SLA agreed with **PDO**; 24/7 coverage applies only where the Gold 24/7 option has been purchased.

PART FIVE

Pricing models, terms, and the path to a signed agreement with **PDO**

32. Commercial Models

The Live Quiz Platform is offered under three distinct commercial models, so that **PDO** can select the ownership, cost, and delivery structure that best fits its procurement approach. Final pricing for each model is set out in the accompanying Commercial Quotation and confirmed after the discovery phase described in Section 24.

32.1 Model A — Annual SaaS Subscription

An annual, subscription-based model in which Altanfith hosts and operates the platform on **PDO's** behalf. Includes: annual licence; cloud hosting; maintenance; security updates; backup; monitoring; support; standard upgrades; unlimited named administrative accounts and unlimited participant access (Section 14), with an agreed number of sessions; an AI usage allowance, with optional overage pricing; customisation priced separately from the base subscription.

32.2 Model B — Perpetual Licence / One-Time Purchase

A one-time purchase model suited to organisations that prefer to own the software outright. Includes: one-time software licence; deployment; initial configuration; training; documentation; an optional source-code licence; optional source-code ownership; optional annual maintenance, recommended as a percentage of software value set out in the Commercial Quotation, payable annually in advance. Major new features are excluded unless separately agreed, and infrastructure costs are excluded unless explicitly listed in the quotation.

32.3 Model C — Enterprise Custom Development Partnership

An ongoing partnership model for organisations requiring deep customisation and integration work. Includes: a discovery workshop; customised workflows; custom reports; system integrations (single sign-on, Microsoft Entra ID / Active Directory, SAP SuccessFactors (or other HR systems via supported integration methods), LMS, Microsoft Teams, Power BI); dedicated support; phased delivery; change requests handled and quoted separately (Section 37); and dedicated development hours or a retainer arrangement.

32.4 Model Comparison

	Model A — SaaS	Model B — Perpetual Licence	Model C — Enterprise Partnership
Ownership	Altanfith-hosted; no source-code ownership by default	Client-owned licence; source code optional	Per agreed scope; typically Altanfith-hosted with client-owned outputs
Upfront cost	Low — annual subscription	Higher — one-time licence fee	Variable — scoped by engagement

Hosting	Altanfith Managed Cloud Infrastructure	Client-managed, on-premises or client-selected cloud	Per final design (Section 18)
Upgrades	Included, standard upgrades	Major features excluded unless agreed	Delivered as part of the ongoing partnership
Best for	Fastest implementation; lowest initial investment	Organisations preferring outright ownership	Organisations needing deep customisation and integration

33. Pricing Approach

This proposal deliberately contains no pricing values. Each of the three commercial models described in Section 32 is priced under the structures set out in the accompanying **Commercial Quotation (ref. ATD-LQP-2026-003)**, which is the sole pricing document for this engagement and should be read together with this proposal. Final pricing for **PDO** is confirmed after the discovery phase (Section 24, Phase 1), once the factors listed in Section 33.1 are known.

33.1 Pricing Methodology

Final pricing for **PDO** depends on a defined set of factors, confirmed during discovery: number of organisations/departments · expected concurrent users · number of live sessions · hosting model · data residency · AI usage · level of customisation · system integrations · support SLA · cybersecurity requirements · source-code requirements · reporting requirements · language requirements · training requirements · project duration.

34. Assumptions

This proposal, and the pricing set out in the accompanying Commercial Quotation (ref. ATD-LQP-2026-003), are prepared on the basis of the following assumptions. Where an assumption does not hold for **PDO**, scope and pricing are revisited during discovery.

1. **PDO** provides timely access to relevant stakeholders throughout the engagement. 2. **PDO** provides branding material (logo, colour palette, style guidance) required for UI customisation. 3. **PDO** provides infrastructure for an on-premises deployment, where selected. 4. **PDO** reviews and approves security and network configurations as they are proposed. 5. Third-party software licences are excluded from pricing unless explicitly listed in the commercial agreement. 6. AI API costs beyond the agreed allowance may be billed separately. 7. Internet connectivity is required to use cloud-based AI generation features. 8. On-premises AI deployment, if pursued, may require separate, dedicated infrastructure. 9. Requirements identified outside the agreed scope are handled through the change-request process (Section 37). 10. Final integration scope depends on the availability of the relevant third-party APIs. 11. **PDO** is responsible for the accuracy of its own content, unless an AI content-review service is separately included in scope. 12. AI-generated content should be reviewed by **PDO** before it is used in a live or published context. 13. Production rollout depends on successful completion of User Acceptance Testing (Section 29).

35. Exclusions

The following items are excluded from this proposal and, unless separately listed in the commercial agreement, are not included in the pricing set out in the accompanying Commercial Quotation.

1. Hardware procurement. 2. **PDO**'s data-centre costs. 3. Third-party software licences. 4. Microsoft licences. 5. SMS charges. 6. Email service charges. 7. AI API overage charges beyond the agreed allowance. 8. External penetration testing (available as an optional service, Section 19). 9. Data migration from unknown or undocumented legacy systems. 10. Unsupported third-party integrations. 11. Major scope changes (handled via the change-request process, Section 37). 12. Custom mobile applications, unless explicitly included in scope. 13. 24/7 support, unless purchased as part of the Gold package (Section 30). 14. Legal or regulatory certification. 15. Cybersecurity certification. 16. Formal compliance accreditation. 17. Travel outside Muscat, unless separately agreed.

36. Risk and Mitigation

The table below sets out the risks Altanfith typically manages on an engagement of this type, and how each is mitigated. Likelihood is confirmed jointly with **PDO** during discovery.

Risk	Likelihood	Impact	Mitigation
Infrastructure readiness	[TBC]	Medium	Early technical assessment (Phase 2); readiness checklist confirmed before deployment
Cybersecurity approval delays	[TBC]	High	Client security team engaged from Phase 3; architecture submitted early for review
Scope changes	[TBC]	Medium	Formal change-request process (Section 37); milestone-based sign-offs
Third-party API limitations	[TBC]	Medium	APIs validated during discovery; fallback approaches designed where needed
AI service availability	[TBC]	Low	AI features degrade gracefully; the core platform operates independently of AI services
User adoption	[TBC]	Medium	Structured training, bilingual materials, and a hypercare period after go-live
Poor venue/participant internet connectivity	[TBC]	Medium	Automatic session reconnection; venue network checklist ahead of live events

High concurrent usage	[TBC]	High	Load testing before go-live; sizing based on the validated baseline of 400+ concurrent participants per live session
Delayed client feedback	[TBC]	Medium	Weekly governance cadence; agreed review SLAs built into the project plan
Data migration complexity	[TBC]	Medium	Migration treated as a scoped, optional phase with its own assessment (Phase 12)
Integration delays	[TBC]	Medium	Integrations phased separately (Phase 8); core go-live is not blocked by integration work
Security testing findings	[TBC]	High	Remediation window built into the plan; retest before go-live (Phase 10)
Internal change management	[TBC]	Medium	Sponsor-led communication plan; departmental change champions

37. Change Request Process

Any requirement outside the agreed scope is handled through a formal change-request process, so that scope, cost, and schedule remain visible and agreed by both parties at every step: a change is requested by either party; Altanfith assesses its impact on scope, cost, and schedule; a written quotation is provided to **PDO**; the change proceeds only after **PDO**'s written approval; and, once approved, delivery is scheduled into the project plan. No change work is carried out without prior written approval.

38. Intellectual Property

Intellectual property ownership is set out clearly and is linked to the commercial model selected (Section 32).

- Standard platform intellectual property remains with Altanfith unless expressly transferred by separate agreement.
- **PDO**'s content and data — including quiz content, participant data, and reports — remain the property of **PDO** at all times (see also Section 21).
- Source-code ownership is not included by default under any commercial model; it applies only where expressly stated in the signed commercial agreement (see Model B, Section 32.2).

39. Proposal Validity

This proposal, including the commercial terms set out in Sections 32, 33, and 40, is valid for a period of **30 days** from **22/07/2026**. Beyond this period, pricing and availability are subject to reconfirmation by Altanfith.

40. Payment Terms

Proposed terms — subject to negotiation. The terms below illustrate the structure Altanfith typically proposes; final payment terms for **PDO** are agreed in the signed commercial agreement.

- Currency: Omani Rial (OMR). Amounts are exclusive of applicable taxes unless otherwise stated.
- Milestone payment structure for project-based engagements: **40%** upon contract signature, **40%** upon User Acceptance Testing (UAT) approval, **20%** upon production go-live. This payment schedule is recommended and remains subject to commercial negotiation and mutual agreement.
- SaaS subscription (Model A): annual payment in advance.
- Maintenance (Model B, optional): annual payment in advance.
- Change requests (Section 37): quoted and invoiced separately.
- Travel outside Muscat: charged separately, where required and agreed in advance.
- Third-party services: charged at cost, plus an agreed administration fee where applicable.

41. Termination and Exit

Either party may terminate the agreement on the notice period set out in the signed contract. On termination, Altanfith provides **PDO** with an export of its data in an agreed format, carries out an orderly handover of the platform and any outstanding documentation, and deletes client data in line with the retention terms agreed in the contract. For on-premises deployments, **PDO's** data never leaves its own environment at any point, including at termination, since it was never hosted outside client control.

42. Future Roadmap

The items below represent Altanfith's future development direction for the Live Quiz Platform. All roadmap items are **Future enhancement** or **Optional**, are not part of the current platform, and their delivery for **PDO** is subject to a separate agreement.

42.1 Learning & Assessment

LMS module · employee certification · training course management · interview assessment module · recruitment assessment · HSE compliance assessments · employee performance evaluation · employee learning profiles · automated certificate generation · gamification, badges & achievements. **Future enhancement**

42.2 AI

AI-generated reports · AI recommendations · chatbot assistant · private, on-premises AI deployment.
Future enhancement

42.3 Integrations

Microsoft Teams · Outlook · Microsoft Entra ID SSO · Active Directory · SAP SuccessFactors (or other HR systems) · Power BI. **Optional** (scoped under Model C, Section 32.3)

42.4 Platform

Mobile application · offline mode · advanced multilingual support · Oman-based hosting. **Future enhancement**

43. Why AI Tanfith AI Dwaliah SPC

Selecting a delivery partner for a platform of this kind is as much about the partner as it is about the software. The reasons set out below are ordered by the value they bring to **PDO**, with our local presence as the foundation that supports the offer, not the offer itself.

An enterprise engagement and assessment platform, already proven in real operational use (Section 11) — not a concept or a demonstration environment. Deep customisation across branding, workflows, reporting, and integrations, rather than a fixed, one-size-fits-all product. In-house AI development capability, already reflected in the platform's current AI-assisted generation (Section 13). Flexible commercial models — SaaS, perpetual licence, and enterprise partnership (Section 32). Cloud, on-premises, and hybrid deployment options (Section 18). A structured support and maintenance model (Section 30). A continuous improvement roadmap (Section 42). A long-term partnership approach to delivery and support. And, underpinning all of it: an Omani company, with local understanding, direct technical support, and fast communication in Arabic and English, without offshore handover delays.

44. Next Steps

We propose the following steps to move from this proposal to a signed agreement and, ultimately, a live platform for **PDO**.

Step	Description
1. Proposal review meeting	Review this proposal together, answer questions, and clarify any assumptions
2. Discovery workshop	Confirm detailed requirements, infrastructure, and integration scope (Section 24, Phase 1)
3. Confirmed scope & pricing	Finalise scope and issue confirmed pricing based on the factors in Section 33.1

4. Contract	Execute the signed commercial agreement, including final terms (Sections 32, 40)
5. Project kickoff	Begin delivery under the methodology set out in Section 24

45. Acceptance Page

This proposal is accepted and agreed by the signatories below on behalf of their respective organisations.

For Petroleum Development Oman (PDO)	For Al Tanfith Al Dwaliah SPC
Name: _____	Name: _____
Title: _____	Title: _____
Signature: _____	Signature: _____
Date: _____	Date: _____

46. Contact Details

Company	Al Tanfith Al Dwaliah SPC
Solution	Live Quiz Platform
Prepared by	Haitham Al Lamki
Email	haitham@altanfith.com
Phone	+968 9937 1775
Country	Sultanate of Oman

47. Appendices

47.1 Appendix A — Glossary

Term	Meaning
API	Application Programming Interface — the interface through which systems exchange data
Hypercare	A period of heightened, close-support monitoring immediately following go-live (Section 24, Phase 16)
JWT	JSON Web Token — a common secure, token-based authentication format
LMS	Learning Management System
MFA	Multi-Factor Authentication
RBAC	Role-Based Access Control — access permissions governed by assigned roles
RLS	Row-Level Security — database-enforced, per-record data isolation
RTL	Right-to-Left — the text layout direction required for Arabic
SLA	Service Level Agreement — defined response and resolution targets for support
SSO	Single Sign-On — authentication against a shared identity provider
TLS	Transport Layer Security — encryption protocol used to secure data in transit
UAT	User Acceptance Testing — client-led testing against agreed acceptance criteria before go-live
WebSocket	A persistent connection protocol used for the platform's real-time features

47.2 Appendix B — Placeholder Index

The following placeholders are used in this proposal and should be completed before contract signature: **[TBC]** (risk likelihood, Section 36), confirmed jointly during discovery. All pricing values are carried exclusively in the accompanying Commercial Quotation (ref. ATD-LQP-2026-003).

47.3 Appendix C — Referenced Documents

The following documents are expected to be referenced by, or read together with, this proposal; the final list is confirmed during discovery:

- Client RFQ / RFP / Statement of Work (where applicable)
- Non-Disclosure Agreement (NDA)
- Project Charter
- Technical Design Document
- Solution Architecture Document
- Security Review Report
- Risk Assessment
- Client Information Security Standards
- Data Classification Policy
- HSE Training Standards (where applicable)